ModelIt!

Teacher's Guide

How Does a Neural Network Learn?

Modeling Artificial Intelligence Training, Weights, and Pattern Recognition

Quick Reference

Grade Level	Grade 11L3
NGSS Standard	HS-ETS1-1, HS-ETS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ETS1-1, HS-ETS1-4

Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use computational models to analyze how mathematical optimization through backpropagation creates emergent pattern recognition capabilities from simple weighted connections.
Disciplinary Core Idea	**ETS1.A Defining and Delimiting Engineering Problems / ETS1.B Developing Possible Solutions** — Students define the engineering problem of creating AI systems that are both accurate and fair, then develop solutions addressing training data, architecture, and evaluation criteria.
Crosscutting Concept	**Systems and System Models** — Students model the AI training pipeline as a system where inputs (data, architecture, hyperparameters) flow through a complex optimization process to produce outputs (predictions, biases, capabilities) with emergent behaviors not predictable from individual components.

CAST Testing Alignment

MULTIPLE CHOICE:

A neural network trained on 10 million medical images achieves 97% accuracy on the test set overall, but only 82% accuracy for images from patients with dark skin tones. Which model variable most likely explains this disparity?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why scaling up a neural network (more parameters, more training data, more compute) does not automatically solve the bias problem. Describe the specific mechanism by which biased training data produces biased predictions regardless of network size.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

In 2012, a neural network called AlexNet won an image recognition competition by a margin so large it shocked the field. In 2016, AlphaGo defeated the world champion at Go — a game many thought would resist AI for decades. In 2022, ChatGPT demonstrated conversational abilities that seemed indistinguishable from human writing. In 2024, AI systems can generate photorealistic images, write code, compose music, and pass medical licensing exams. All of this emerged from the same fundamental mechanism: adjusting connection weights in neural networks through exposure to massive amounts of data.

How Neural Networks Work

A neural network is a series of layers, each containing nodes (artificial neurons) connected to nodes in the next layer. Each connection has a weight — a number that determines how strongly one node influences the next. An input (like an image's pixel values) enters the first layer, gets multiplied by weights, passed through activation functions that add non-linearity, and propagated through successive layers until the final layer produces an output (like 'cat' or 'dog'). Initially, all weights are random and the network's predictions are meaningless. Training is the process of systematically adjusting these weights so that correct predictions are amplified and incorrect ones are suppressed.

Backpropagation and Learning

Backpropagation is the algorithm that makes neural network training possible. After each prediction, the loss function calculates the error — how far the prediction was from the correct answer. Backpropagation then traces this error backward through the network, computing how much each individual weight contributed to the mistake. Weights that pushed toward the wrong answer are decreased; weights that pushed toward the right answer are increased. The learning rate controls how large each adjustment is. Over millions of training iterations across thousands or millions of examples, the weights gradually encode the statistical patterns in the data.

The Bias Amplification Problem

Neural networks learn whatever patterns exist in their training data — including patterns of discrimination. If historical healthcare data shows that Black patients received less aggressive treatment for the same symptoms (which is documented fact), an AI trained on this data will learn to recommend less aggressive treatment for Black patients, not because the AI is racist but because it faithfully reproduces the patterns in its training data. Worse, AI systems can amplify biases: a hiring AI trained on a company's historical hiring data (which skewed male) actively filtered out female candidates. The data encodes the past; the AI projects that past into the future.

The Scale Paradigm and Its Limits

The dominant paradigm in modern AI is scaling: larger models trained on more data with more compute produce better results. GPT-3 had 175 billion parameters trained on 300 billion tokens. GPT-4 is estimated at over 1 trillion parameters. Each generation costs roughly 10 times more to train — GPT-4's training likely exceeded $100 million in compute costs alone. Scaling has produced remarkable capabilities, but returns are diminishing logarithmically while costs increase exponentially. A model 10 times larger may only be 10-20% better on benchmarks. This suggests that scaling alone will not achieve artificial general intelligence — something qualitatively different may be required.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven key components of a neural network training system: training data volume, training data quality, network depth, learning rate, prediction accuracy, overfitting risk, and computational cost.

E **Establish Relationships**

Students map the relationships showing that training data quality is the foundational variable affecting all outcomes, that scaling (data volume and network depth) provides diminishing returns at exponentially increasing cost, and that the gap between training and validation accuracy reveals overfitting.

V **Visualize & Model**

Students build a computational model showing the seven-component AI training system with competing pressures between accuracy, fairness, cost, and generalization.

E **Evaluate Outcomes**

Students run balanced training, scale-up, and biased data scenarios to discover how each parameter affects the system and identify the conditions under which AI systems become both accurate and fair.

R **Revise & Extend**

Students add regularization strength, inference latency, or catastrophic forgetting to model more realistic AI system dynamics and limitations.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"An AI system can now pass the bar exam, diagnose diseases from X-rays, write poetry, and generate photorealistic faces of people who do not exist. All of this comes from the same mechanism: adjusting numbers in a massive web of connections. Today you find out how — and what is really happening under the hood."*

Watch For

Open with demonstrations: show an AI-generated image, an AI-written poem, and an AI medical diagnosis. Ask: how did a machine learn to do all of this? What does it actually understand?

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the engine behind the AI revolution — and discovering that what looks like intelligence is actually sophisticated pattern matching on an enormous scale."*

Watch For

Have students read objectives. Pre-teach 'backpropagation' as the learning algorithm and 'overfitting' as the enemy of real learning. Use the test-memorizer analogy.

Slide 3: Big Question

3 min

Say: *"The answer might surprise you: it does not understand any of it. It finds statistical patterns in data. The question is whether pattern matching at extreme scale starts to look like understanding — and whether the difference matters."*

Watch For

Quick-write: Students write what they think 'understanding' means and whether a machine can truly understand. Share out. This philosophical question becomes concrete through the model.

Slide 4: LEVER Framework

2 min

Say: *"We are modeling a system where the quality of what goes IN determines the quality and fairness of what comes OUT — and where bigger is not always better."*

Watch For

Preview LEVER steps. Emphasize that this model directly connects to AI systems students use every day — social media algorithms, recommendation systems, ChatGPT.

Slide 5: Activity 1: Components

8 min

Say: *"Seven components. Four inputs you can control: data volume, data quality, network depth, and learning rate. Three outputs that emerge from training: accuracy, overfitting, and cost. Watch what happens when you change the inputs."*

Watch For

Guide through all seven components. Discuss why data volume, quality, depth, and learning rate are external — engineers choose these. Accuracy, overfitting, and cost are consequences.

Slide 6: Activity 2: Connections

8 min

Say: *"Data quality is the keystone. You can build the most brilliant architecture and train for months on the most powerful hardware — but if the data is biased, the AI will be biased. Garbage in, garbage out, at billion-dollar scale."*

Watch For

Help students map relationships. Highlight the asymmetry: poor data quality undermines everything, but good data quality does not guarantee good outcomes without adequate architecture.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us test: can you find settings where the AI is both accurate AND fair? What happens when you scale everything up? And what happens when the data is biased?"*

Watch For

Students predict outcomes before each scenario. The biased data scenario is the critical learning moment — watch accuracy diverge between population groups even with a well-designed network.

Slide 8: Discoveries

5 min

Say: *"The biggest insight: AI does not learn truth from data. It learns patterns from data. If the data contains injustice, the AI learns injustice and calls it optimization."*

Watch For

Lead discussion connecting model findings to real AI controversies: facial recognition accuracy disparities by race, hiring algorithms that discriminated, healthcare AI that underserved Black patients.

Slide 9: STEM Challenge

12 min

Say: *"A hospital wants to deploy AI to save lives — but the AI works better for some patients than others. Your job: fix it before anyone gets hurt."*

Watch For

Groups design fair AI training protocols for the healthcare scenario. Must specify data collection changes, architecture decisions, fairness metrics, and monitoring plans. Present with model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Training Data Volume, Training Data Quality, Network Depth, Learning Rate
- **INTERNAL:** Prediction Accuracy, Overfitting Risk, Computational Cost

Relationships

- **Training Data Quality to Prediction Accuracy:** POSITIVE (+) — Higher quality training data — more accurate labels, more diverse representation, fewer systematic biases — directly produces a model that makes more accurate predictions across all populations because it has learned genuine patterns rather than artifacts of poor data curation.
- **Network Depth to Overfitting Risk:** POSITIVE (+) — Deeper networks with more layers and parameters have more capacity to memorize specific training examples rather than learning generalizable patterns. Without sufficient regularization and data, increased depth increases the gap between training accuracy and real-world performance.
- **Training Data Volume to Computational Cost:** POSITIVE (+) — More training data requires proportionally more computation to process — each training epoch must iterate through every example, and larger datasets require more storage, memory, and GPU hours, increasing total training cost.
- **Training Data Quality to Overfitting Risk:** NEGATIVE (-) — Higher quality, more diverse training data reduces overfitting risk because the model encounters more varied examples that force it to learn general patterns rather than memorizing specific quirks of a narrow, unrepresentative dataset.

Key Discoveries

- Neural networks do not understand anything in the human sense — they are sophisticated pattern matching systems that find statistical correlations in training data, and their apparent intelligence emerges from the scale and structure of these correlations
- Training data quality is the most critical variable for real-world impact — a brilliantly designed network trained on biased data will produce biased outputs that can reinforce and amplify societal discrimination at scale
- Scaling up (more data, more parameters, more compute) has produced remarkable capability gains, but with exponentially diminishing returns and exponentially increasing costs — each new generation of AI models costs roughly 10 times more to train
- The gap between training accuracy and real-world performance (overfitting) reveals a fundamental limitation — networks can appear to understand while merely having memorized patterns that fail in novel situations

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"AI systems are objective because they are based on math, not human judgment"	Neural networks are trained on data created by humans in a society with systemic biases. The math is objective; the data is not. When historical data reflects discrimination — less aggressive medical treatment for Black patients, fewer women hired into tech roles, heavier policing in communities of color — the AI learns these patterns and reproduces them as optimal predictions. The AI does not know it is being unfair; it is optimizing the objective function it was given using the data it was trained on. The bias is not in the algorithm — it is in the data, which reflects the world.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Bigger AI models are always better"	The scaling paradigm has produced impressive results, but the relationship between scale and capability is logarithmic, not linear. Doubling the parameter count of a model from 500 billion to 1 trillion might improve accuracy by 2-5%, while increasing training cost from $50 million to $100+ million. The model shows diminishing returns clearly: the accuracy curve flattens while the cost curve steepens. Furthermore, larger models trained on more data simply amplify whatever patterns are in that data — including biases — at greater scale and with more confidence.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Neural networks think and understand like human brains"	Despite the name 'neural network,' the connection to biological brains is superficial. Biological neurons use electrochemical signaling, have complex internal dynamics, form and dissolve connections based on experience, and operate in a biochemical soup that influences processing. Artificial neurons multiply inputs by weights and apply simple mathematical functions. Artificial neural networks have no goals, no experiences, no consciousness, and no understanding of what their outputs mean. They are extraordinarily powerful pattern-matching machines, but equating their processing with human cognition is a category error.